MATH 214 – Assignment 1
Root-Finding: Bisection, Fixed-Point, and Newton

1 Algorithms and Implementation

The three solvers are implemented in C++17 as modular functions (header `solvers.hpp`). Each accepts a `std::function` for the user's f (and, where required, f' or g), an initial guess or bracket, $\varepsilon = 10^{-6}$, and a maximum iteration cap of 100. Each returns a `SolverResult` struct containing the final root x_n, residual $|f(x_n)|$, total iteration count, the full sequence $\{x_0, x_1, \ldots, x_n\}$, a convergence flag, and a textual termination reason.

The stopping criteria are applied in this order on every iteration:

1. residual test — $|f(x_k)| < \varepsilon$,
2. step test — $|x_{k+1} - x_k| < \varepsilon$,
3. iteration cap — $k = 100$.

For Bisection the step test uses the distance between consecutive midpoints; for Fixed-Point the residual test is computed externally against the original f for diagnostic purposes (the iteration itself only needs g). The Newton routine additionally aborts safely if $f'(x_k) = 0$ or if any iterate becomes non-finite (overflow/NaN).

2 Test Problems and Setup

2.1 Problem 1: General Polynomial $f(x) = x^3 - x - 2$

Reference root: $x^* \approx 1.5213797068045676$.

- **Bisection** on $[1, 2]$: $f(1) = -2$, $f(2) = 4$, sign change present.
- **Fixed-Point, form A**: $g_A(x) = (x + 2)^{1/3}$. $g'_A(x^*) = \frac{1}{3}(x^* + 2)^{-2/3} \approx 0.144 < 1 \Rightarrow$ contraction near x^*, so iteration converges.
- **Fixed-Point, form B**: $g_B(x) = x^3 - 2$. $g'_B(x^*) = 3(x^*)^2 \approx 6.94 > 1 \Rightarrow$ *not* a contraction; iteration diverges.
- **Newton** from $x_0 = 1.5$ with $f'(x) = 3x^2 - 1$.

2.2 Problem 2: Transcendental $f(x) = \cos(x) - x$

Reference root: $x^* \approx 0.7390851332151607$ (the Dottie number). Tested Newton from a wide range of starting points $x_0 \in \{0, 0.5, 1, 5, 50\}$ to expose how the basin of attraction behaves. Bisection on $[0, 1]$ included as a baseline.

2.3 Problem 3: Multiple Root $f(x) = x^3$

The root $x^* = 0$ has multiplicity $m = 3$, so $f'(x^*) = f''(x^*) = 0$. Standard Newton's quadratic-convergence theorem assumes a simple root and therefore fails to apply. Tested:

- Bisection on $[-1, 1]$ (the first midpoint $c = 0$ is the exact root, so termination is immediate);

- standard Newton from $x_0 \in \{1.0,\, 0.5\}$;
- modified Newton with the multiplicity correction $x_{k+1} = x_k - m\, f(x_k)/f'(x_k)$ from $x_0 = 1.0$.

3 Results

Table 1: Summary of all runs. Tolerance $\varepsilon = 10^{-6}$, max_iter = 100.

Problem	**Method / setup**	**Iter.**	**Final root**	**Residual**
P1	Bisection on $[1, 2]$	20	1.521380425	4.27×10^{-6}
P1	Fixed-Point, $g(x) = (x+2)^{1/3}$, $x_0 = 1.5$	7	1.521379679	1.64×10^{-7}
P1	Fixed-Point, $g(x) = x^3 - 2$, $x_0 = 1.5$	10	diverged $(-\infty)$	∞
P1	Newton, $x_0 = 1.5$	2	1.521379806	5.89×10^{-7}
P2	Bisection on $[0, 1]$	20	0.739085197	1.08×10^{-7}
P2	Newton, $x_0 = 0.0$	4	0.739085133	2.85×10^{-10}
P2	Newton, $x_0 = 0.5$	3	0.739085134	1.18×10^{-9}
P2	Newton, $x_0 = 1.0$	3	0.739085133	2.85×10^{-10}
P2	Newton, $x_0 = 5.0$	20	0.739085145	1.93×10^{-8}
P2	Newton, $x_0 = 50.0$	35	0.739085315	3.04×10^{-7}
P3	Bisection on $[-1, 1]$	1	0	0
P3	Newton (standard), $x_0 = 1.0$	12	7.71×10^{-3}	4.58×10^{-7}
P3	Newton (standard), $x_0 = 0.5$	10	8.67×10^{-3}	6.52×10^{-7}
P3	Newton (modified, $m = 3$), $x_0 = 1.0$	1	0	0

3.1 Convergence Plots

Figures 1–3 are semilogarithmic plots of $|x_k - x^*|$ versus iteration index k. On a $\log_{10}$ scale, *linear* convergence appears as a straight line and *quadratic* convergence appears as a curve whose slope steepens with each step (the number of correct digits roughly doubles). All curves are drawn in black; distinct line styles and markers identify the series.

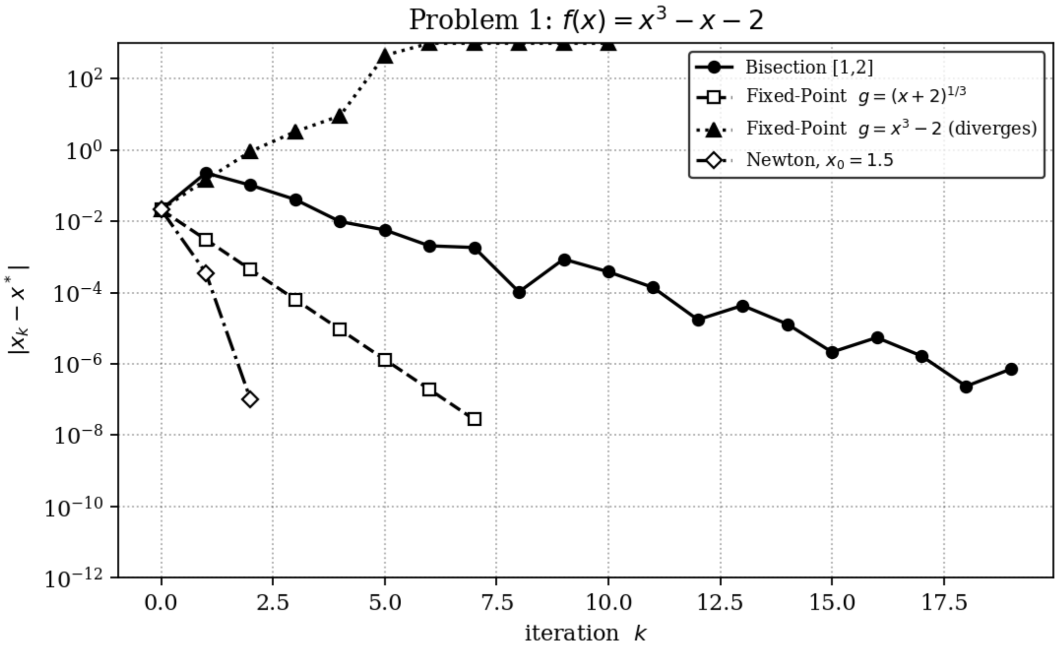

Figure 1: **Problem 1.** Newton (dash-dot, diamonds) drops in 2 steps; fixed-point with $g_A = (x+2)^{1/3}$ (dashed, squares) is straight on the semilog axes, i.e. linear; bisection (solid, circles) decays roughly like $(1/2)^k$; the bad fixed-point $g_B = x^3 - 2$ (dotted, triangles) blows up off the top of the plot.

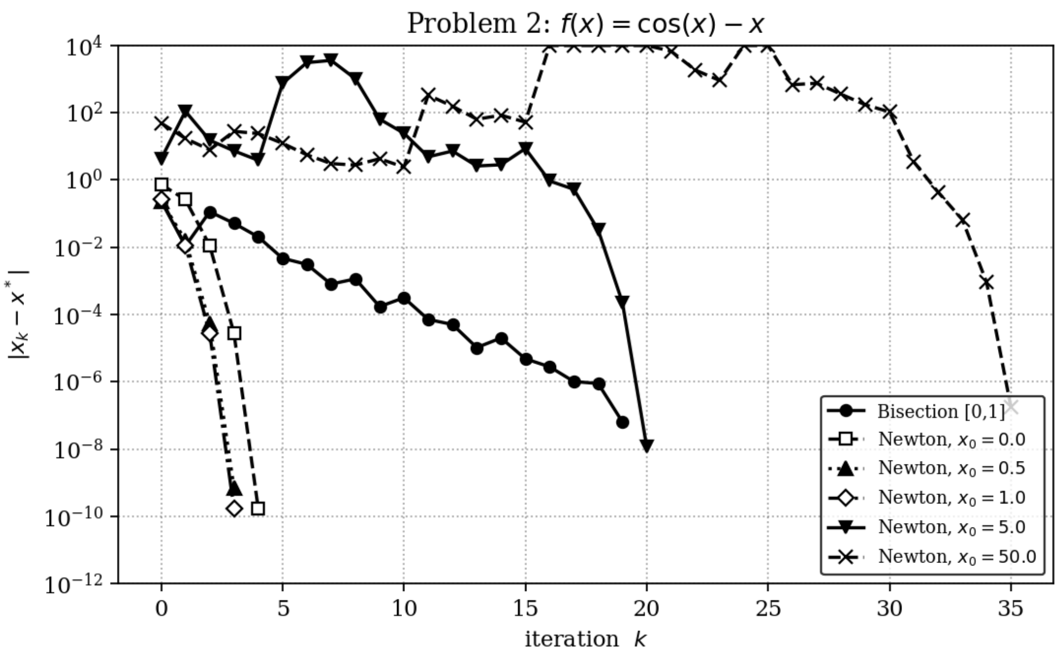

Figure 2: **Problem 2.** For $x_0 \in \{0, 0.5, 1\}$ Newton converges in 3–4 iterations with the characteristic accelerating drop of quadratic convergence. With $x_0 = 5$ and $x_0 = 50$ the iterate is thrown far outside the basin: $|x_k - x^*|$ peaks at $\approx 3.6 \times 10^3$ (iter 7) for $x_0 = 5$ and at $\approx 3.0 \times 10^6$ (iter 16) for $x_0 = 50$. The plot is clipped at 10^4 to keep the convergent series legible, so excursions above this line are flat; the actual iterates wander chaotically until one happens to land inside the basin of attraction, after which standard quadratic convergence resumes (the steep drop into 10^{-7} at the right end of each curve).

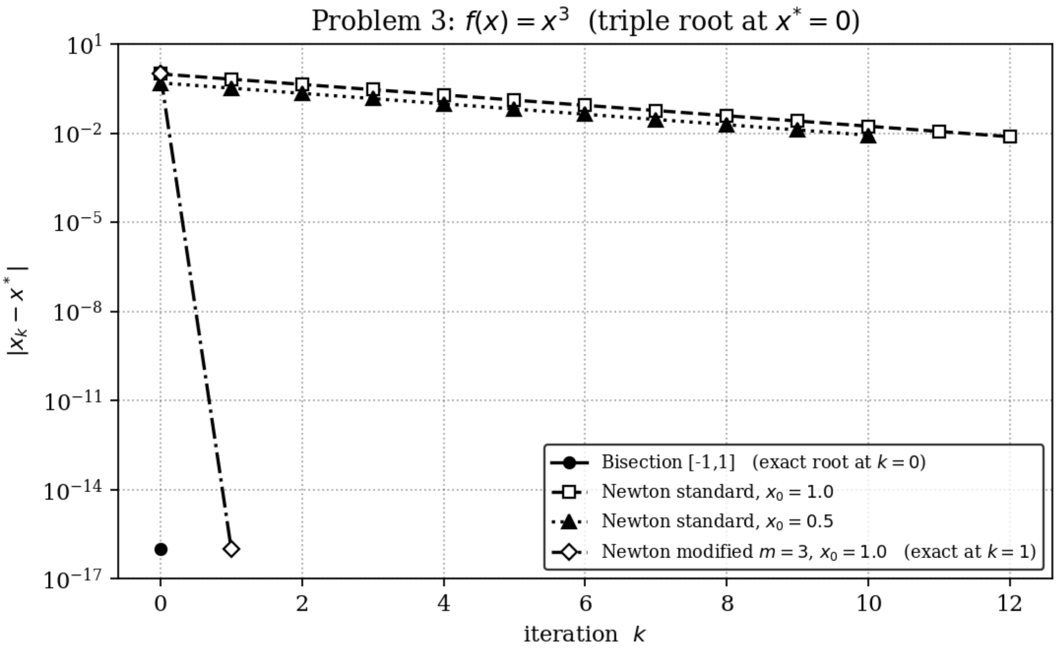

Figure 3: **Problem 3.** Standard Newton on $f(x) = x^3$ produces a perfectly straight line on semilog axes — the signature of *linear* convergence with rate 2/3 (see §4). The modified Newton iteration with $m = 3$ (the steep vertical drop on the left) reaches the exact root in a single step, restoring quadratic behavior.

4 Discussion

4.1 Robustness vs. Speed

Most robust: *Bisection.* Provided the initial interval contains a sign change, it is mathematically guaranteed to converge — it cannot diverge, oscillate, or hit a zero derivative. Its weakness is rate: error roughly halves each step, giving fixed linear convergence with rate $1/2$ (about $\log_2 10 \approx 3.3$ iterations per decimal digit).

Fastest: *Newton's method,* when applicable. For Problems 1 and 2 it converged in 2–4 iterations from reasonable starting points, versus 20 for Bisection at the same tolerance. The price is fragility: Newton needs f', can be sabotaged by zero or small derivatives, can throw the iterate to a different basin (or to infinity), and degrades to linear convergence at multiple roots. Fixed-point sits between the two: cheap (no derivative needed), but its convergence depends entirely on the algebraic choice of g.

4.2 Why Newton or Fixed-Point Slowed Down or Failed

Fixed-point with $g_B(x) = x^3 - 2$ (Problem 1). Convergence of $x_{k+1} = g(x_k)$ near a fixed point x^* requires $|g'(x^*)| < 1$. Here $g'_B(x^*) = 3(x^*)^2 \approx 6.94$, so the iteration locally *magnifies* every error by nearly a factor of 7 per step. Starting at $x_0 = 1.5$ near the root, errors blow up super-exponentially and the iterate overflows to $-\infty$ within 10 steps. Form $g_A(x) = (x+2)^{1/3}$ has $|g'_A(x^*)| \approx 0.144$, giving linear convergence with that contraction ratio — visible as a line of slope $\log_{10}(0.144) \approx -0.84$ per iteration in Figure 1.

Newton from far starting points (Problem 2). For $f(x) = \cos x - x$, $f'(x) = -\sin x - 1 \in [-2, 0]$, vanishing at $x = \pi/2 + 2\pi n$. When $x_0 = 5$ or $x_0 = 50$, several early Newton steps land near such near-singular points where $|f'| \ll 1$, producing a huge update $-f/f'$ that flings the iterate to a distant region. Indeed, the recorded iterates reach $|x| \approx 3.6 \times 10^3$ for the $x_0 = 5$ run and $|x| \approx 3.0 \times 10^6$ for the $x_0 = 50$ run during their wandering phases. The iterate eventually — by chance — lands inside the basin of attraction near $x^* \approx 0.739$, after which the classical quadratic drop reappears. This is the textbook "bad initial guess" failure mode: not formal divergence, but global behavior that can be wildly far from the eventual root, with a quadratic finish only once the iterate stumbles into a good neighborhood. Bisection, by contrast, would have converged in 20 iterations no matter what bracket of unit length we chose.

Newton at the triple root (Problem 3). For $f(x) = x^3$ the Newton step simplifies analytically:

$$x_{k+1} = x_k - \frac{x_k^3}{3x_k^2} = x_k - \frac{x_k}{3} = \frac{2}{3}\, x_k.$$

Thus $|x_k - x^*| = (2/3)^k |x_0|$ exactly — linear convergence with rate $2/3$. To reach $\varepsilon = 10^{-6}$ from $x_0 = 1$ requires roughly $\lceil 6 \ln 10 / \ln(3/2) \rceil \approx 34$ iterations to satisfy the *step* test. The runs terminate earlier (at $k = 12$ from $x_0 = 1$) because the *residual* test $|x_k^3| < 10^{-6}$ is much easier to satisfy than the step test — $x \approx 0.0077$ already gives $x^3 \approx 4.6 \times 10^{-7}$. The reported root is therefore not six-digit accurate; this is a known and important pitfall when the residual test is used at high-multiplicity roots, since f is extremely flat there. The general theoretical reason is that Newton's quadratic convergence proof requires $f'(x^*) \neq 0$. At a root of multiplicity m the local rate degrades to linear with asymptotic constant $(m-1)/m$; with $m = 3$ that gives $2/3$, matching observation exactly. The simple fix used in our modified Newton, $x_{k+1} = x_k - m\, f/f'$, restores quadratic convergence; with $m = 3$ on $f(x) = x^3$ we get $x_{k+1} = x_k - 3 \cdot x_k/3 = 0$, which is why it lands exactly in one step.

4.3 Theoretical vs. Observed Convergence Rates

- **Bisection (linear, rate $1/2$).** On Problems 1 and 2 the method needed 20 iterations to bring the interval width down to $\varepsilon = 10^{-6}$ from a unit interval. Theoretically $(1/2)^{20} =$

$9.5 \times 10^{-7} < 10^{-6}$, in exact agreement.

- **Fixed-Point with g_A on P1 (linear, rate ≈ 0.144).** Predicted reduction per step is $\log_{10}(0.144) \approx -0.84$ decades; the slope of the dashed series in Figure 1 matches this within rounding.
- **Newton on P1 and P2 with good starts (quadratic).** The error-vs-iter curves are visibly concave on the semilog plot — the gap between consecutive points roughly doubles — and the iteration terminates in 2–4 steps, consistent with the quadratic bound $|e_{k+1}| \leq C|e_k|^2$.
- **Newton on P3 (linear, rate $2/3$).** The dashed and dotted lines in Figure 3 are essentially straight with slope $\log_{10}(2/3) \approx -0.176$ per iteration — exactly what theory predicts for a root of multiplicity 3.

Bottom line. Choose Bisection when reliability matters and a sign-change bracket is available; choose Newton when a good initial guess and a cheap derivative are available and the root is simple; reach for Fixed-Point only after analytically checking $|g'(x^*)| < 1$ for the chosen rearrangement. None of the three is universally best, and Problem 3 illustrates that even Newton's celebrated quadratic rate is not a guarantee — it is a theorem with hypotheses.